CTF Writeup: The Reason for the Photograph

OSINT & Geospatial Intelligence (GEOINT) Challenge Analysis • Final Flag:
hiCTF{zona_arqueologica_de_iglesia_vieja_squeaked.flinches.overlarge}

Challenge Overview Establishes the Spatial Riddle and Flag Structure

Attribute	Details
Challenge Title	The Reason for the Photograph (OSINT / GEOINT)
Provided Archive	`the-reason-for-the-photograph.zip` (SHA-256: `186ac4022dd75e5538446758d0b8b2d26bac21362002a9b1b0c1aa2bce995d79`)
Extracted Image	`place.jpg` (`1456 × 720` lossless RGB PNG screenshot, stripped of EXIF and GPS metadata)
Flag Format	`hiCTF{site_name_three.word.address}`
Solved Flag	`hiCTF{zona_arqueologica_de_iglesia_vieja_squeaked.flinches.overlarge}`

> *"Why frame a clearing when the ruins are right here?" "Because anyone can take a picture of what they are looking at. I wanted a picture from where I stood." "So the true attraction is at your back?" "Exactly. What mattered wasn't in front of the lens."*
> *They christened this sanctuary for a faith it never served, mistaking primordial megaliths for a house of prayer. Time preserved the lie in its name, but the mountain remembers. Find where the observer stood. Look away from the clearing.*



Figure 1 (Original Artifact): The uncropped `1456 × 720` challenge image (`place.jpg`) showing the open grassy plaza clearing, the charred leaning tree trunk in the foreground, the circular stone ring on the shaded ground to the left, and the distant stepped megalithic granite wall across the lawn.

Step 1: Archive Inspection and Visual Forensics Confirm a 360° Photo Sphere Capture

Unpacking `the-reason-for-the-photograph.zip` produces a single image named `place.jpg`. Inspecting the binary header reveals that the file is actually a lossless `1456 × 720` PNG screen capture (~2:1 aspect ratio) with no EXIF tags, camera serial numbers, or embedded GPS coordinates:

```
sha256sum the-reason-for-the-photograph.zip
# 186ac4022dd75e5538446758d0b8b2d26bac21362002a9b1b0c1aa2bce995d79
unzip the-reason-for-the-photograph.zip && file place.jpg
# place.jpg: PNG image data, 1456 x 720, 8-bit/color RGB, non-interlaced
```

Performing visual forensics on the image isolates four distinct geometric and archaeological markers that narrow down both the capture mechanism and the physical setting:

1. **Rectilinear 360° Projection Distortion:** Severe radial stretching in the upper-left and upper-right tree canopy corners confirms that `place.jpg` is a rectilinear viewport screenshot taken inside an equirectangular 360° Google Street View Photo Sphere rather than a standard physical camera lens.
2. **Circular Stone Altar Ring:** On the shaded ground to the left of center lies a low circular stone ring constructed of dressed granite blocks on the plaza perimeter.
3. **Distant Megalithic Platform and Central Ramp:** Across the sunlit grassy clearing stands a long, stepped pre-Hispanic structure built of massive granite slabs, bisected by a lighter grey central stone ramp with a dark boulder resting in front of its base.
4. **Charred Leaning Tree Trunk:** A prominent dark, leaning tree trunk angles from the bottom-right foreground toward the upper-left center, serving as an exact azimuth reference inside the 360° panorama.

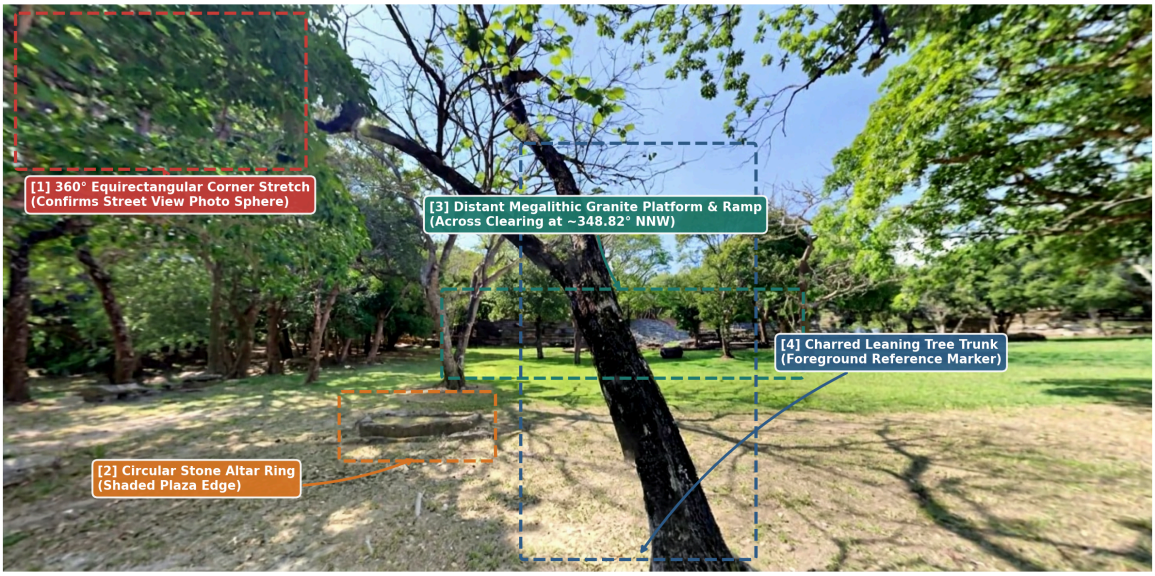


Figure 2 (Forensic Callout Map): Annotated breakdown of `place.jpg` highlighting `[1]` the 360° equirectangular projection stretch in the upper canopy, `[2]` the circular stone altar ring, `[3]` the distant megalithic granite platform and ramp across the clearing, and `[4]` the leaning tree trunk in the foreground.



Figure 3 (High-Resolution Photographic Macro Crops): Close-up zooms extracted from `place.jpg` isolating (A) the distant megalithic granite wall, central ramp, and dark foreground boulder across the clearing, (B) the circular carved stone ring on the shaded plaza floor, and (C) the western stone retaining blocks along the forest perimeter.

Step 2: Historical Riddle Analysis Identifies Zona Arqueológica de Iglesia Vieja

The second paragraph of the prompt encodes the exact archaeological and etymological backstory of the site:

- **"Primordial megaliths":** Points to a pre-Hispanic Mesoamerican site built using monumental megalithic granite masonry rather than typical limestone or stucco (Mesoweb — Akira Kaneko, 2011).
- **"They christened this sanctuary for a faith it never served, mistaking primordial megaliths for a house of prayer. Time preserved the lie in its name, but the mountain remembers.":** Identifies Zona Arqueológica de Iglesia Vieja ("Old Church"), located 4 km north of Tonalá, Chiapas, Mexico, atop **Cerro de la Cruz** in the Sierra Madre de Chiapas (Wikipedia — Zona arqueológica de Iglesia Vieja). Built by the Zoque civilization between 400 BC and 600 AD, its massive stepped granite platforms were later mistaken by local inhabitants for the ruins of a colonial Christian church (*Iglesia Vieja*).

Formatting the full official Google Maps and INAH designation (**Zona Arqueológica de Iglesia Vieja**) into lowercase ASCII snake_case produces the required `site_name` component: `zona_arqueologica_de_iglesia_vieja`.



Figure 4 (The Sanctuary Behind the Lens): Architectural view of the monumental Zoque stepped granite masonry and central ramp at Zona Arqueológica de Iglesia Vieja — the primary attraction standing directly at the observer's back while the camera framed the open clearing.

Step 3: Reversing the 360° Photo Sphere Bearing to "Look Away from the Clearing"

Searching the cropped megalithic wall from Figure 3A matches a Google Street View 360° Photo Sphere at Zona Arqueológica de Iglesia Vieja (`panoid: CAoSLEFGMVFpcE1Rckd1aGpTaUlBbmI4U0xFdkFDTjRYN2xISzdTeDJCeU1lWlZI`), pinned on Google Maps at **`16.1297539, -93.7451699`** (`///amazes.communes.hairier`).

The riddle's dialogue explains why the raw Google Maps pin coordinate is one grid square off from the final answer: the screenshot `place.jpg` is framed at a compass heading of **`348.82°` (North-Northwest)** looking out across the clearing. Rotating **180° to `168.82°` (South-Southeast — "Look away from the clearing")** turns directly toward the megalithic ruin at the observer's back. Stepping one 3m × 3m What3Words grid square South-East along that `168.82°` bearing moves from the pin (`16.1297539, -93.7451699`) to **`16.129730, -93.745132`**.

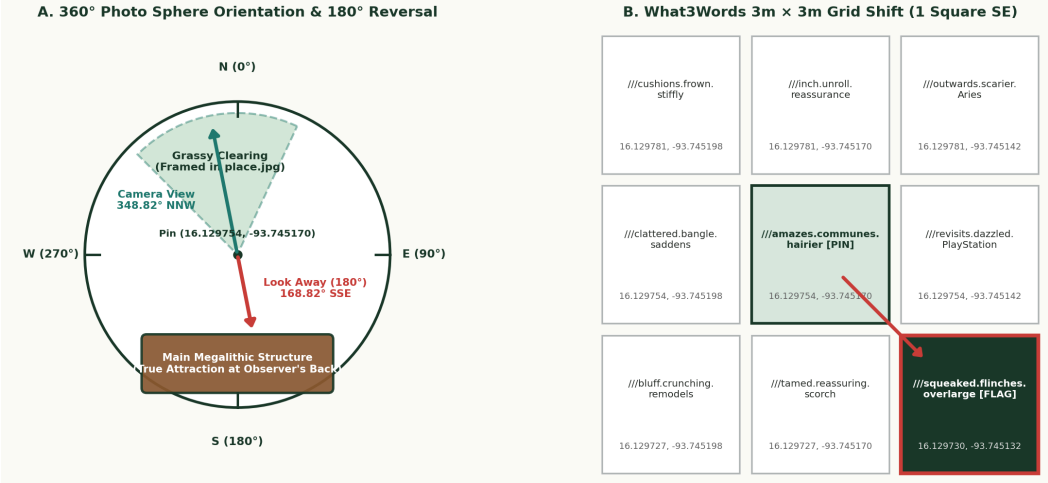


Figure 5 (Geospatial Bearing & What3Words Grid Shift): (A) 360° compass orientation showing the `348.82°` NNW clearing view versus the 180° reversal (`168.82°` SSE) toward the megalithic structure at the observer's back. (B) The 3m × 3m What3Words grid around `///amazes.communes.hairier` showing the 1-square South-East step to `///squeaked.flinches.overlarge`.

Step 4: What3Words Grid Verification Produces the Final Flag

Converting the surrounding 3m × 3m squares via Grid Reference Finder confirms the exact What3Words address at the observer's South-East offset:

Grid Position / Role	WGS84 Coordinates (`Lat, Long`)	Offset from Pin	What3Words Address (Grid Reference Finder)
Google Maps Photo Sphere Pin	`16.1297539, -93.7451699`	`0m N, 0m E` (Center)	`///amazes.communes.hairier`
1 Square South (`180°`)	`16.1297269, -93.7451699`	`3m S, 0m E`	`///tamed.reassuring.scorch`
1 Square East (`90°`)	`16.1297539, -93.7451419`	`0m S, 3m E`	`///revisits.dazzled.PlayStation`
Observer Spot (~`168°` SE, Away from Clearing)	**`16.1297300, -93.7451320`**	**`3m S, 3m E` (Target)**	**`///squeaked.flinches.overlarge`**

Combining the normalized site name (`zona_arqueologica_de_iglesia_vieja`) with the resolved 3-word address (`squeaked.flinches.overlarge`) completes the flag: **`hiCTF{zona_arqueologica_de_iglesia_vieja_squeaked.flinches.overlarge}`**.